

Rutgers University-Camden
Department of Public Policy and Administration
Faculty Peer Evaluation of Classroom Teaching

Course:

- 56:219:521 DATA VISUALIZATION (dat sci)
- 56:824:728 DATA VISUALIZATION (soc sci/pub pol)
- 56:834:653 DATA VISUALIZATION (soc sci/pub pol)

Instructor: Adam Okulicz-Kozaryn **Date Evaluated:** April 3, 2025

Note: Given the long time elapsed since the class observation, I supplemented my notes by watching the class video.

1. Subject of Class:

1) visualization techniques for data clustering, e.g. dendrograms and k-means scatterplots, 2) annotation of graphs using matplotlib, 3) “flipped classroom” to respond to student questions about a problem set.

2. Relevance of the subject to the course:

The content was appropriate to a course on data visualization, touching on important methods for representing clusters and relationships in data.

3. A. Was an assignment made: Yes _____ No ☒

Although no new assignment was introduced during the session, students appeared engaged with an assignment from the prior week and came prepared to discuss it.

B. Were students prepared: Yes ☒ No _____

4. Clarity of the ideas presented:

The description of the class as given to the students:

[apr3] clustering

a bit similar to previous scatterplots; clustering is about grouping / finding patterns, and very much data sciency!

we only scratch the surface here--just few popular vis for clustering;
i don't cover here the method, algorithms etc

Adam worked systematically through the material, alternating between code demonstrations in an online notebook and examples. The examples were presented clearly, but the presentation would have benefited from stronger substantive and conceptual framing. For example, the class could have explored more explicitly when to use a dendrogram versus a scatterplot with k-means clustering. For another, an “elbow plot” was presented as diagnostic for k-means clustering, but there was no explanation of what the plot represented, only that it tells you then ideal number of groups (based on what?). Without more grounding, the different types of visualizations are just black boxes.

While links to more in-depth discussion of the visualizations in relevant textbooks were provided, they lacked specific guidance (e.g., chapters or sections), which may make it difficult for students to locate the relevant material.

5. Manner of delivery:

Adam put students at ease and was very responsive to student questions. He made good use of examples, both simple examples using a small dataset followed by more complete examples using graphics from published papers.

6. Effective use of teaching aide(s), if they apply:

Rather than a whiteboard or PowerPoint slides, Adam used an online coding notebook, an environment for writing and executing code. This is an appropriate format given the material and the emphasis on coding. However, the notebook’s structure—one long, cumulative document covering several classes—made it difficult to follow at times, as his frequent scrolling up and down through the long document disrupted the flow. Segmenting the material into separate notebooks for each class or adding clearer section headings could improve clarity and pacing.

7. Student participation: The students were engaged throughout and asked appropriate questions. Adam had a good rapport with the students. When students sought deeper conceptual explanations (e.g., how clustering algorithms work), the responses could have gone beyond general descriptions to help students better understand the mechanics and assumptions of these techniques.

8: Other comments:

The syllabus and class notes would benefit from more careful organization and editing. Clearer structure, proofreading, and a consistent format would strengthen the professional presentation of the course materials and better reflect the high standards of our program.

More broadly, the course could be enhanced by integrating theoretical and design-oriented perspectives on data visualization—topics such as perception, cognition, and design principles. These form the intellectual foundation of the field and help students make informed decisions about how to represent data effectively and truthfully.

Adam’s course provides strong coding content and fosters an open, interactive learning

environment. The main opportunities for improvement lie in strengthening conceptual foundations, improving the organization and clarity of course materials, and ensuring that design and communication principles are integrated alongside technical skills. These adjustments would make the class more rigorous and balanced, aligning it more closely with the best practices in data visualization pedagogy. Without that, students may learn *how* to make various chart but not *why* certain choices communicate data more truthfully or effectively.

Paul A. Jorgensen

Visitor: _____

Date: 11/11/2025 Instructor: Bob Olin-Kay Date: _____
nov21 2025

Acceptance or comments:

The evaluation was helpful as it made me realize that my html bullet points didn't render and thus obscured readability; and it got me thinking about spending less time on code as AI takes over. Other than that the evaluation was unhelpful. I would like to raise several issues so that we can do better in the future.

Paul does not teach Python or data visualization. Neither does he use Python in his research. I was arguing to have someone else evaluate my teaching. I even found a replacement to do evaluation. Then I can't get the evaluation for 7 months (class was in April, I get evaluation in November) despite multiple emails and reminders. And then evaluation is mostly unhelpful as outlined below with my inline responses.

"the presentation would have benefited from stronger substantive and conceptual framing. "

It's a data visualization class, not math or algorithms class.

"an "elbow plot" was presented as diagnostic for k-means clustering, but there was no explanation of what the plot represented, only that it tells you then ideal number of groups (based on what?). Without more grounding, the different types of visualizations are just black boxes."

That's the whole point of the "elbow plot" to tell the number of groups. Again, it's a data visualization class, not math or algorithms class.

"While links to more in-depth discussion of the visualizations in relevant textbooks were provided, they lacked specific guidance (e.g., chapters or sections), which may make it difficult for students to locate the relevant material."

No. There is a reference to a specific chapter: "and especially machine learning ch:
<https://jakevdp.github.io/PythonDataScienceHandbook/05.00-machine-learning.html>"

"the notebook's structure—one long, cumulative document covering several classes—made it difficult to follow at times, as his frequent scrolling up and down through the long document disrupted the flow. Segmenting the

material into separate notebooks for each class or adding clearer section headings could improve clarity and pacing."

There are not only section headings, but there are 3 levels of them: chapters, sections, and sub-sections (and I even police this in students' assignments—I take points off if students don't use sections).

"Clearer structure, proofreading, and a consistent format would strengthen the professional presentation of the course materials and better reflect the high standards of our program."

I was trying to discuss and get a sense of what Paul means here but I wasn't able to get detailed or specific information. As far as I can tell it's about writing style: Paul is traditional and formal. And I'm quite the opposite: modern, straightforward, and informal.

My materials have a clear structure and consistent format. I actually have a specific philosophy here to be straightforward and informal. I follow Zinnser's "On Writing Well: The Essential Guide to Mastering Nonfiction Writing and Effective Communication." For instance: "Never say anything in writing that you wouldn't comfortably say in conversation. If you're not a person who says "indeed" or "moreover," or who calls someone an individual ("he's a fine individual"), please don't write it." I was asking Paul for reference to his style, but I didn't get any.

I even wrote a paper on cluttered and ornamented grandiose prose in social science as compared to natural science: <https://link.springer.com/article/10.1007/s11192-012-0937-9>

So called social science "professional presentation" in writing can typically actually obscure meaning and communication. Again, following Zinnser, the goal should be simplicity and readability.

Per "high standards of our program"--i think my teaching has a good standard, e.g., my current students publish in good journals: https://theaok.github.io/okulicz_kozaryn_cv.pdf#page=6

And, what do we mean by "high standards of our program"? Is it formal traditional writing? Does such writing translate into desired outcomes, i.e. communication and learning? (especially in methods/coding classes). These are ever more relevant questions as we are increasingly over-burdened by information. Clarity, simplicity, and readability seem a way to go.

"More broadly, the course could be enhanced by integrating theoretical and design-oriented perspectives on data visualization—topics such as perception, cognition, and design principles. These form the intellectual foundation of the field and help students make informed decisions about how to represent data effectively and truthfully. Adam's course provides strong coding content and fosters an open, interactive learning environment. The main opportunities for improvement lie in strengthening conceptual foundations, improving the organization and clarity of course materials, and ensuring that design and communication principles are integrated alongside technical skills. These adjustments would make the class more rigorous and balanced, aligning it more closely with the best practices in data visualization pedagogy. Without that, students may learn how to make various chart but not why certain choices communicate data more truthfully or effectively."

These were covered in the course earlier. The class was geared towards the students, not towards the visitation, i.e. the visited Apr3 class was not self-contained, it did not include important elements included in other classes.

I did reach out to Paul to discuss, build bridges, and understand each other better. We had 2 discussions, somewhat helpful, but still I was not able to get enough details and specificity. I got a sense that it is time to wrap up, hence my lengthy response above. For the future, I am hoping we can agree to disagree and work out constructive compromises.

The evaluation was also helpful in a sense that I watched the class video and noted I could have had a little more eye contact and walk around more and be less on the monitor. But again as outlined above most of the evaluation is unhelpful--I hope in the future we can be more constructive. I am excited to discuss and collaborate on teaching. I think we could focus more on teaching.

Rev. 1/12